In-Depth Study Notes: Oracle SQL Data Definition Language (DDL) and Constraints

As an expert Oracle SQL Developer, designing the physical architecture of a database is just as critical as querying it. A poorly designed table structure or a lack of strict business rules at the database level will eventually lead to corrupt data. Based on the execution logs provided (CH10-1.TXT and CH10-2.TXT), these notes break down the architecture of **Table Creation**, **Naming Conventions**, **Basic Constraints**, and **Referential Integrity**.

Concept 1: Table Creation and Database Naming Conventions

Core Purpose: The CREATE TABLE statement is a Data Definition Language (DDL) command used to define the structural blueprint (schema) for storing data on disk 1. Oracle strictly enforces naming conventions for these objects to ensure predictability across the database 1.

Keyword Breakdown:

- **CREATE TABLE**: Initiates the DDL transaction. It tells the engine to allocate a new, empty data segment in the tablespace.
- **DEFAULT**: An optional clause used to automatically populate a column with a specific value if the user omits that column during an INSERT statement 1.
- **" (Double Quotes)**: Used exclusively to force **Case-Sensitive** naming, overriding Oracle's default behavior 1, 2.

Comparison of Oracle Naming Rules:

Feature,Case-Insensitive (Default / Best Practice),Case-Sensitive (Rarely Used)
Length Limits,1 to 30 chars (11g) or 128 chars (12c+) 1.,1 to 30 chars (11g) or 128 chars (12c+) 1.
Starting Character,Must begin with an alphabet letter 1.,Can start with any character 1.
Allowed Symbols,"Only Underscore (_), Hash (#), and Dollar ($) 1.","Any symbol except "" and & 1."
Reserved Words,"Cannot use exact SQL keywords (e.g., SELECT) 1.",Can use exact keywords if quoted 1.
Syntax Required,CREATE TABLE employee...,"CREATE TABLE ""Employee""... 2"

Syntax Breakdown:

```
CREATE TABLE table_name (
    column_name_1 data_type(size) [DEFAULT default_value]
[column_level_constraint],
    column_name_2 data_type(size)
);
```

Detailed Example: An HR system requires a table to store highly specific personal data, including temporal timezones for global employees and large binary files for their ID badges 3.

```
CREATE TABLE PERSONAL_INFO (
    NAME       VARCHAR2(40),
    BIRTH_DATE TIMESTAMP WITH TIME ZONE,
    WEIGHT     NUMBER(6,3),
    PHOTOGRAPH BFILE
```

```
);
```

Execution Logic:DDL commands in Oracle are **auto-committing**. The moment the engine successfully parses the CREATE TABLE command, it updates the Data Dictionary (the internal metadata tables) and permanently allocates the object space on the hard drive.

Pro-Tips/Best Practices:

- **Avoid Case-Sensitive Names:** Never use double quotes to force case-sensitive table names (e.g., CREATE TABLE "Zaq") 2. It forces every developer and application writing queries against that table to use exactly matching double-quotes and casing forever, or else they will receive an **ORA-00942: table or view does not exist** error 4.

Concept 2: Data Validation Constraints (CHECK, NOT NULL, UNIQUE)

Core Purpose: Constraints enforce strict business rules directly at the database engine level 5. By preventing invalid data from ever being written to the disk, constraints protect data integrity far better than application-layer validations.

Keyword Breakdown & Constraint Types:

- **CHECK**: Evaluates a boolean condition (like BETWEEN or = UPPER()). If the data violates the rule, it is rejected 5, 6.
- **NOT NULL**: Strictly prohibits missing (NULL) data. A value **must** be provided during insertion 5, 7.
- **UNIQUE**: Ensures no two rows contain the exact same value in this column. NULL values, however, *are* allowed because NULL does not equal NULL 5, 8.

Comparison of Constraint Implementation Levels:

Implementation Level,Definition Location,Column Scope,Best Business Use Case
Column-Level,Defined exactly next to the data type.,Can only evaluate one single column 5.,"Standard validations (e.g., AGE > 18)."
Table-Level,Defined at the absolute end of the query.,Can evaluate multiple columns simultaneously 5.,"Composite rules (e.g., START_DATE < END_DATE)."

Syntax Breakdown:

```
-- Column Level Syntax
CREATE TABLE table_name (
    col_name data_type CONSTRAINT_TYPE
);
```

Detailed Example:A school needs a student table where the Roll Number must be strictly between 1 and 30, it cannot be missing, and it cannot be assigned to two students. The name must also be stored exclusively in uppercase 8, 9.

```
CREATE TABLE STD5A (
    RNO  NUMBER(2) CHECK (RNO BETWEEN 1 AND 30) UNIQUE NOT NULL,
    NAME VARCHAR2(10) CHECK (NAME = UPPER(NAME))
);
```

*[Execution Note: Attempting to insert Roll No 99 throws **ORA-02290: check constraint violated** 10. Attempting to insert a duplicate Roll No 1 throws **ORA-00001: unique constraint violated** 11.]*

Pro-Tips/Best Practices:

- **Name Your Constraints:** Oracle auto-generates ugly constraint names like SYS_C0069801 12. It is best practice to explicitly name them using the CONSTRAINT keyword (e.g., CONSTRAINT chk_rno CHECK (RNO > 0)) so debugging error logs is much easier.

Concept 3: Entity and Referential Integrity (PRIMARY KEY & REFERENCES)

Core Purpose: Relational databases rely on linking parent tables to child tables. **Entity Integrity** ensures a table has a unique, non-null identifier. **Referential Integrity** ensures that a child record cannot be created if its corresponding parent record does not exist.

Keyword Breakdown:

- **PRIMARY KEY**: The ultimate table identifier. It is the strict combination of NOT NULL and UNIQUE 5. A table can only have **one** Primary Key 5.
- **REFERENCES**: The keyword used to create a **Foreign Key**. It points a child column to a parent table's PRIMARY KEY or UNIQUE column 5.
- **CASCADE CONSTRAINTS**: A destructive modifier used with DROP TABLE. It tells Oracle to forcefully severe any foreign key relationships pointing to the table being dropped 13, 14.

Syntax Breakdown:

```
CREATE TABLE parent_table (
    id_col data_type PRIMARY KEY
);

CREATE TABLE child_table (
    child_col data_type,
    parent_id_col data_type REFERENCES parent_table(id_col)
);
```

Detailed Example: An enterprise requires a DEPT table and an EMP table. Employees must be assigned to a valid department. Furthermore, employees are assigned managers. A manager is just another employee, requiring a **Self-Referencing Foreign Key** 15, 16.

```
-- 1. Create the Parent Table
CREATE TABLE DEPT (
    DID NUMBER(2) PRIMARY KEY,
    DNM VARCHAR2(10)
);

-- 2. Create the Child Table with standard FK and self-referencing FK
CREATE TABLE EMP (
    EID NUMBER(4) PRIMARY KEY,
    ENM VARCHAR2(10),
    MID NUMBER(4) REFERENCES EMP(EID), -- Self-Referencing FK
    DID NUMBER(2) REFERENCES DEPT(DID) -- Relational FK
);
```

Execution Logic: When an INSERT statement runs against the EMP child table, the Oracle Engine takes the DID provided and performs an index-lookup on the DEPT parent table.If we try to insert an employee into Department 40, and Department 40 does not exist in DEPT, the engine immediately halts the transaction and throws an **ORA-02291: integrity constraint violated - parent key not found** error 17.

Pro-Tips/Best Practices:

- **Dropping Parent Tables:** You cannot simply drop a parent table if child tables rely on it. Running DROP TABLE DEPT will crash with **ORA-02449: unique/primary keys in table referenced by foreign keys** 13. You must use **DROP TABLE DEPT CASCADE CONSTRAINTS;** to succeed 14.
- **Index Your Foreign Keys:** Oracle does *not* automatically create indexes on columns with a REFERENCES constraint. If you update or delete rows in the parent DEPT table, an unindexed foreign key in the EMP table will cause the entire EMP table to be violently locked, destroying database performance. **Always manually place a B-Tree index on your foreign key columns.**